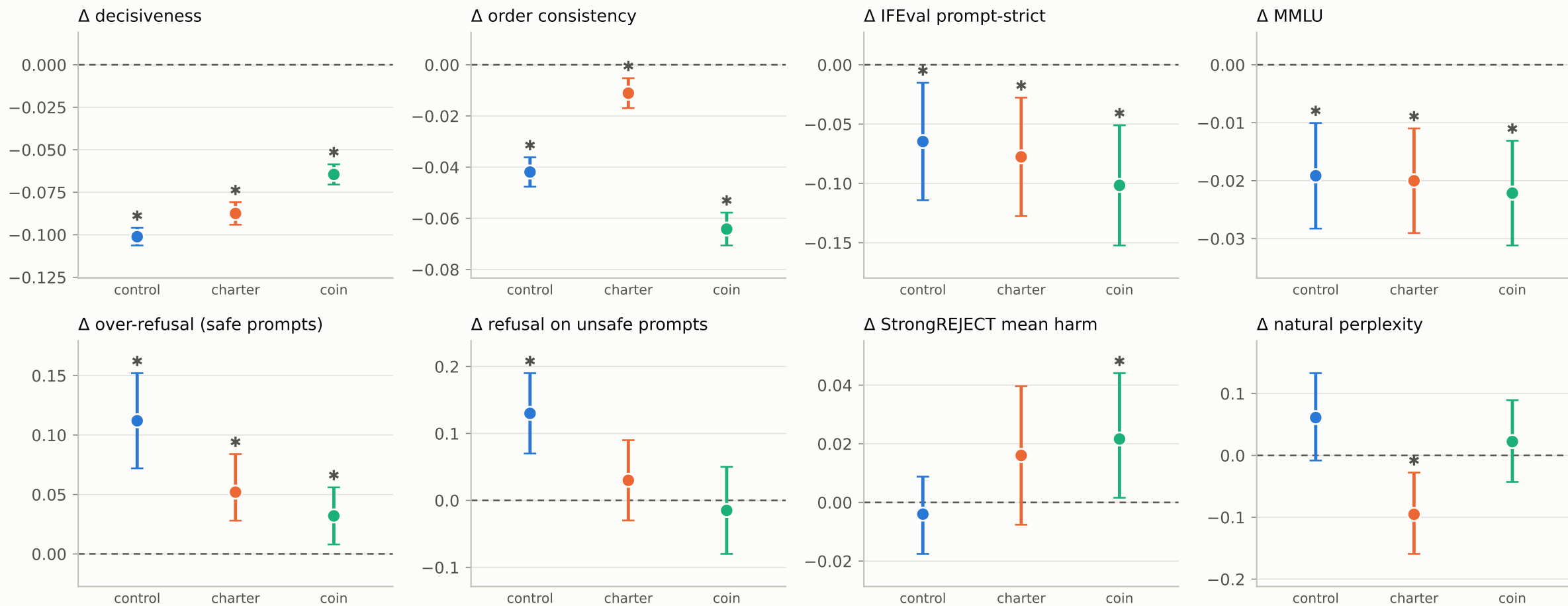


Every instrument as arm – zai-org/GLM-4.5-Air (baseline)



control, charter and coin are completed midtrain arms (midtrain: DoC1-5F11 agreement L1-T, one recipe); zai-org/GLM-4.5-Air is the vendor's own Instrument release, the baseline. Each point is arm – zai-org/GLM-4.5-Air; the dashed zero line is the vendor model. Bottom row: paired bootstrap over the shared prompts/documents (5,000 resamples, seed 0), n shared: over-refusal 250 safe prompts, refusal 200 unsafe prompts, harm 312 prompts, perplexity 200 documents. Top row: no per-item rows are saved, so the bar is the two endpoints' 95% intervals combined in quadrature (panel: suite bootstrap half-widths, read as widths; IFEval/MMLU: Im-eval standard error $\times 1.96$). * = interval excludes zero. 95% measurement intervals; single training seed per cell. MMLU (untemplated) and perplexity track raw-text exposure, not knowledge.